

Operator Discovery Report: descriptor_scaffold_selector_v1

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Choose a simple baseline scaffold deterministically from instance descriptors, prioritizing bottleneck and trap-aware heuristics, with a robust fallback for mixed/unknown structure.
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:clustered_local_search:random_restart_2opt.
- Surviving candidate: False.
- Problem class where it helps: clustered_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: b8844c7280347b68.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on heldout_tsplib with gap delta 0.005953.
- Underperforms on elongated_corridor_tsp with gap delta 0.034174.
- Underperforms on grid_like_tsp with gap delta 0.170114.
- Underperforms on nearest_neighbor_trap_tsp with gap delta 0.3987.
- Underperforms on two_cluster_bottleneck_tsp with gap delta 0.022965.
- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into nearest_neighbor_2opt.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Does not transplant cleanly into random_restart_2opt.
- Does not transplant cleanly into sparse_three_opt.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.
- Runtime inflation is high at 24.116978.

Transplant Results

- nearest_neighbor_2opt gap delta 0.005953, runtime delta 3562.927486, distance-eval delta 6175991.0.
- cheapest_insertion_2opt gap delta 0.113275, runtime delta 123.355086, distance-eval delta -585067.0.
- random_restart_2opt gap delta 0.026279, runtime delta 3334.782043, distance-eval delta 6161039.714286.
- sparse_three_opt gap delta 0.007239, runtime delta 3542.660543, distance-eval delta 6172569.714286.
- clustered_local_search gap delta -0.014371, runtime delta 3448.417014, distance-eval delta 6164819.571429.

Instance-Family Results

- heldout_tsplib operator gap 0.241011 vs baseline 0.235058.
- clustered_tsp operator gap 0.0 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.037378 vs baseline 0.003204.
- grid_like_tsp operator gap 0.27681 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.3987 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.132022 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: 0.057846.
- Runtime inflation: 24.116978.
- Distance-evaluation inflation: 628.593581.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.057846.
- flattened_selector mean gap 0.121317 and delta vs full operator -0.057846.
- shuffled_selector mean gap 0.106492 and delta vs full operator -0.072672.